Diophantine quadruple congruent mod k

Problèmes ouverts — A. Dujella · 2026

Problem 4.5 (Dujella, 2026). Is there any Diophantine quadruple $\{a, b, c, d\}$ such that $a \equiv b \equiv c \equiv d \pmod{k}$ with $k \geqslant 3$?

Remark. It was shown by Dujella and Saradha (Indag. Math. (N.S.) **25** (2014), 131–136) that a positive answer to Problem 4.6 would imply a negative answer to Problem 4.5.